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MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
REGULAR EXAMINATION March 2026
Fourth SEMESTER
Value Addition Course (VAC) - MG4VACCSC200 - GREEN COMPUTING TECHNIQUES (2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer all questions
Each question carries **2** marks

1. **EDI** enables the **electronic exchange of business documents between organizations in a standardized digital format**, reducing paper use and improving efficiency.

2. Green standards provide **guidelines, benchmarks, and environmental regulations** that help organizations reduce energy use, minimize pollution, and ensure environmental compliance.

3. Documentation in **ISO 14000** records **environmental policies, procedures, monitoring results, and compliance activities**, ensuring accountability and continuous environmental improvement.

4. Two technological dimensions supporting Green IT implementation are:

   Energy-efficient hardware and devices
   Virtualization and cloud computing technologies

5. **CSR – Corporate Social Responsibility**

6. **Biodegradable packaging** decomposes naturally by microorganisms and causes less environmental pollution, whereas **non-biodegradable packaging** does not decompose easily and remains in the environment for a long period.

7. Green computing reduces paper and office cost by **promoting electronic communication, digital documentation, and paperless workflows**, thereby minimizing printing and storage expenses.

8. **Green Supply Chain Management (GSCM)** is the integration of **environmental thinking into supply chain activities such as product design, material sourcing, manufacturing, distribution, and disposal**.

9. **Server consolidation** is the practice of **reducing the number of physical servers by combining workloads on fewer servers using virtualization technologies to save energy and resources**.

10. **Handheld computers** allow **digital data entry, storage, and real-time processing**, while **traditional clipboards** rely on **paper-based manual recording and later data transfer**.

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

11. Major problems associated with excessive paper usage

Excessive paper usage leads to several environmental and operational problems. Large quantities of paper production contribute to **deforestation**, which destroys natural ecosystems. The manufacturing process consumes **significant energy and water resources**, increasing environmental impact. Paper waste also increases the **volume of solid waste** in landfills. Organizations incur **high costs for printing, storage, and document management**. In addition, transportation and disposal of paper contribute to **carbon emissions**, affecting environmental sustainability.

12. Features of a Green Data Center

A green data center focuses on **energy efficiency and reduced environmental impact**. It uses **energy-efficient servers and processors** to minimize power consumption. Advanced **cooling systems** such as efficient airflow management reduce energy used for temperature control. **Server consolidation and virtualization** reduce the number of physical machines required. Green data centers often use **renewable energy sources** like solar or wind power. They also implement **power monitoring and energy management systems** to track and optimize energy usage.

13. Application of Kyoto Protocol provisions to address global warming

The **Kyoto Protocol** is an international agreement designed to reduce **greenhouse gas emissions** responsible for global warming. It sets **legally binding emission reduction targets for developed countries**. The protocol introduces mechanisms such as **carbon trading**, allowing countries to buy or sell emission allowances. It also includes the **Clean Development Mechanism (CDM)**, enabling developed countries to invest in emission reduction projects in developing countries. These measures encourage the adoption of **clean technologies and renewable energy**, helping reduce global warming.

14. Importance of employee and stakeholder engagement in environmental practices

Employee and stakeholder engagement is essential for the success of environmental initiatives. Employees help implement **green practices such as energy conservation and waste reduction** in daily operations. Stakeholder involvement ensures **support from management, suppliers, and customers** for sustainability initiatives. Engagement increases **environmental awareness and responsibility** within the organization. It also improves **corporate reputation and transparency**, encouraging organizations to maintain environmentally responsible operations.

15. Major phases in Green Information System development and deployment

Green Information System development involves several phases. The first phase is **requirement analysis**, where environmental objectives and sustainability goals are identified. The second phase is **system design**, focusing on energy-efficient architecture and environmentally friendly technologies. The next phase is **system development**, where software and hardware solutions are implemented. This is followed by **testing and evaluation** to measure environmental performance. Finally, the system is **deployed and monitored**, ensuring continuous improvement and efficient resource usage.

16. Difference between RoHS and EPEAT

RoHS (Restriction of Hazardous Substances) is a regulatory directive that restricts the use of **toxic substances such as lead, mercury, and cadmium** in electronic equipment manufacturing. Its primary focus is on **reducing hazardous materials in electronic products**.

EPEAT (Electronic Product Environmental Assessment Tool) is an **environmental rating system** that evaluates electronic products based on several environmental criteria such as **energy efficiency, recyclability, and environmental design**. While

RoHS mainly addresses hazardous materials, EPEAT provides a **broader assessment of overall environmental performance** of electronic products.

17. Concept and classification of carbon footprint related to IT infrastructure

A **carbon footprint** refers to the **total amount of greenhouse gas emissions generated directly or indirectly by activities related to IT infrastructure**. These emissions occur through the **operation of computers, servers, and data centers**. Carbon footprints can be classified into **direct emissions**, produced by on-site energy use and equipment operation; **indirect emissions**, generated from electricity consumed by IT systems; and **supply chain emissions**, which arise from manufacturing, transportation, and disposal of IT equipment.

18. Converting a conventional PC into a Green PC

A conventional PC can be transformed into a **Green PC** by adopting several green computing practices. Power management features should be enabled to reduce energy consumption when the system is idle. Energy-efficient hardware components such as **Energy Star certified devices** can be used. Replacing **CRT monitors with LCD or LED displays** significantly reduces power consumption. Unnecessary background processes should be minimized to optimize system performance and energy usage. Proper **recycling and reuse of electronic components** also contributes to environmentally sustainable computing practices.